



Databases and Applications

Unit III

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Computer Applications

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Unit III

Normalization: 1NF, 2NF, Functional Dependencies

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Normalization – Introduction

- ▶ Normalization is a process that improves database design by generating relations of higher normal forms.
 - Normal forms discussed: 1NF, 2NF, 3NF, BCNF
 - Objective: ensure every dependency is on the key, the whole key, and nothing but the key
 - Reduces redundancy and simplifies update semantics



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Hierarchy of Normal Forms

- ▶ Normal forms follow a sequence:
 - 1NF is weakest
 - 2NF stronger than 1NF
 - 3NF stronger than 2NF
 - BCNF strongest
- ▶ Any relation in BCNF is also in 3NF, in 2NF, and in 1NF.



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Why Normalization Matters

- ▶ Higher normal forms reduce redundancy and prevent integrity problems.
- ▶ Benefits include:
 - Simpler update semantics
 - Lower redundancy
 - More consistent relational structures



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Functional Dependencies – Definition

- ▶ A functional dependency $A \rightarrow B$ means:
 - If two records have the same A-value, they must have the same B-value
 - A uniquely determines B
- ▶ Example:
 - $\text{EmpNum} \rightarrow \text{EmpEmail}$



- Functional Dependencies from the employee relation:

EmpNum	EmpEmail	EmpFname	EmpLname
123	jdoe@abc.com	John	Doe
456	psmith@abc.com	Peter	Smith
555	alee1@abc.com	Alan	Lee
633	pdoe@abc.com	Peter	Doe
787	alee2@abc.com	Alan	Lee



- ▶ If EmpNum is the primary key, then:
 - $\text{EmpNum} \rightarrow \text{EmpEmail}$
 - $\text{EmpNum} \rightarrow \text{EmpFname}$
 - $\text{EmpNum} \rightarrow \text{EmpLname}$
- ▶ Each employee has one email, firstname, lastname determined by EmpNum.



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Determinants

- ▶ In the FD $\text{EmpNum} \rightarrow \text{EmpEmail}$:
 - EmpNum is the **determinant**
 - Determinant = attribute on the LHS of an FD



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Types of Dependencies

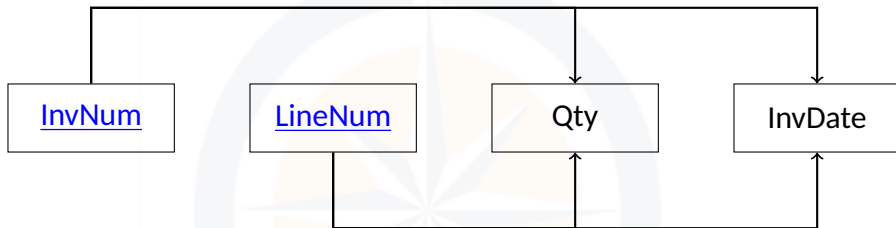
- ▶ Normalization uses:
 - **Partial dependency** – attribute depends on part of a composite key
 - **Transitive dependency** – $A \rightarrow B$ and $B \rightarrow C$ implies $A \rightarrow C$
- ▶ These violations determine which normal form is achieved.



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Partial Dependency – Definition

- ▶ A partial dependency exists when:
 - B depends on A
 - A is part of a composite candidate key
- ▶ Only occurs when candidate keys have multiple attributes.



- ▶ Candidate key: (InvNum, LineNum)
- ▶ InvDate depends only on InvNum.
- ▶ Therefore:
 - InvDate is partially dependent on (InvNum, LineNum)
 - Violates 2NF



1NF – Definition

- ▶ A relation is in 1NF if:
 - All attribute values are single-valued and atomic
 - No repeating groups
 - No multivalued attributes



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1NF – Example of Violation

- The following relation is **not in 1NF** due to multivalued attributes:

EmpNum	EmpPhone	EmpDegrees
123	233-9876	–
333	233-1231	BA, BSc, PhD
679	233-1231	BSc, MSc

- Issues:
- EmpDegrees has multiple values for a single employee
 - Violates atomicity and 1NF requirements



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1NF – Example of Violation

- ▶ Example of not being in 1NF:
 - EmpDegrees = BA, BSc, PhD (multi-valued)
 - Employee 679: BSc, MSc
- ▶ Multi-valued fields violate atomicity.



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Converting to 1NF

► Solution:

- Create separate rows for each degree
- Split into two relations:
 - Employee(EmpNum, EmpPhone)
 - EmployeeDegree(EmpNum, EmpDegree)



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Converting to 1NF

- ▶ To achieve 1NF, multi-valued attributes must be separated into atomic values.
- ▶ Step 1: Create a base **Employee** relation with atomic attributes.

Employee Table

EmpNum	EmpPhone
123	233-9876
333	233-1231
679	233-1231

- ▶ This removes the multi-valued EmpDegrees attribute from the main relation.



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Converting to 1NF

- ▶ Step 2: Create a separate **EmployeeDegree** relation to store all degree values atomically.

EmployeeDegree Table

EmpNum	EmpDegree
333	BA
333	BSc
333	PhD
679	BSc
679	MSc

- ▶ These two tables, when joined, reconstruct the original data without violating 1NF.



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2NF – Definition

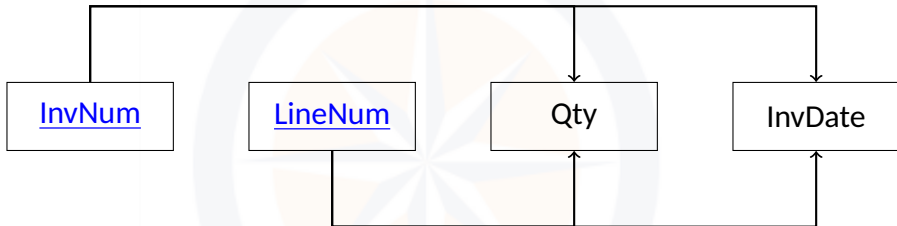
- ▶ A relation is in 2NF if:
 - It is already in 1NF
 - Every non-key attribute is fully dependent on each candidate key
 - No partial dependencies exist



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2NF – Example Analysis

- Consider the relation shown below:



- Issues:

- Partial dependency: InvDate depends only on InvNum
- Composite key attributes InvNum and LineNum
- Repeated invoice dates cause redundancy
- Violates 2NF



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2NF – Example Analysis

- ▶ Relation: InvLine(InvNum, LineNum, ProdNum, Qty, InvDate)
- ▶ Issues:
 - InvDate depends only on InvNum → partial dependency
 - Repeated InvDate across invoice lines → redundancy



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2NF – Decomposition

- ▶ Decompose into two relations:
 - InvLine(InvNum, LineNum, ProdNum, Qty)
 - Invoice(InvNum, InvDate)
- ▶ Removes partial dependency and redundancy.



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2NF – Decomposition

We can *improve* the database by decomposing the relation into two relations:

Relation 1

<u>InvNum</u>	<u>LineNum</u>	ProdNum	Qty
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Relation 2

<u>InvNum</u>	InvDate
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Thank You

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